Data Analytics Using SQL

Question Bank with Answers: Data Exploration

Part A - 2 Marks

Q1. What is data exploration (Exploratory Data Analysis)? Name two tools used for it.

Introduction: Data exploration, also called Exploratory Data Analysis (EDA), is the initial step of the analysis process in which large datasets are examined to understand their main characteristics before formal modelling begins.

- **Purpose:** It uses statistical techniques and visualisation to uncover patterns, relationships, outliers and data quality problems early, so that later modelling and business decisions rest on reliable data.
- **Tools:** Python, R and Tableau are commonly used to draw histograms, scatter plots and box plots. In a database setting, SQL queries using COUNT, AVG and GROUP BY give a quick first summary.

Q2. Differentiate between a predictor variable and a target variable with an example.

Introduction: A predictor (input) variable is used to explain or predict an outcome, whereas the target (output) variable is the outcome that the analysis tries to predict.

Basis	Predictor Variable	Target Variable
Role	Input (independent) variable that supplies information to the model.	Output (dependent) variable whose value the model predicts.
Example	In a student table, Gender, Prev_Exam_Marks, Height and Weight are predictors.	Play_Cricket (1 or 0) is the target when we predict whether a student plays cricket.

Q3. Differentiate between continuous and categorical variables with one example each.

Introduction: A continuous variable takes measurable numeric values over a range, while a categorical variable takes a limited set of labels that place records into groups.

Basis	Continuous Variable	Categorical Variable
Nature and example	Numeric and measurable, for example Height = 178 cm or Salary = 45000.	Labels or groups, for example Gender = M/F or Play_Cricket = 1/0.
Univariate analysis	Mean, median, IQR and standard deviation, shown with a histogram or box plot.	Frequency table with Count and Count%, shown with a bar chart.

Q4. What is an outlier? Give one example.

Introduction: An outlier is an observation that lies far away from the overall pattern of the other values in a sample.

- **Meaning:** Outliers distort statistics such as the mean and standard deviation, so they need close attention before modelling; otherwise they can produce wildly wrong estimates.
- **Example:** In a Salary table where most employees earn between 30000 and 60000, a record of 600000 (perhaps a typing error with an extra zero) stands far above the rest and appears as an outlier on a box plot.

Q5. Write an SQL query to display CustomerID and Name of the customers from the Customer table whose Income value is missing.

Introduction: A missing value in a SQL table is stored as NULL, which means the value is unknown. NULL cannot be compared using =, so the IS NULL operator must be used.

- **Query:** The WHERE clause keeps only those rows in which Income is NULL, and the SELECT list displays the two required columns.

```
SELECT CustomerID, Name
FROM   Customer
WHERE  Income IS NULL;
```

- **Expected output:** Only customers with an unknown Income are listed. For example:

```
CustomerID | Name
-----------+-------
C104       | Ravi
C109       | Meena
```

Q6. Write an SQL query to display the five highest Amount orders from the Orders table so that possible outliers can be inspected.

Introduction: Sorting a numeric column in descending order brings the most extreme values to the top, which is a quick SQL method of spotting possible outliers.

- **Query:** ORDER BY Amount DESC arranges the orders from largest to smallest and LIMIT 5 keeps only the first five rows (use SELECT TOP 5 in SQL Server).

```
SELECT OrderID, CustomerID, Amount
FROM   Orders
ORDER  BY Amount DESC
LIMIT  5;
```

- **Expected output:** One order is far larger than the remaining ones, which marks it as a candidate outlier to be checked.

```
OrderID | CustomerID | Amount
--------+------------+-------
ORD012  | C017       | 9500
ORD004  | C002       | 450
ORD008  | C011       | 440
ORD006  | C005       | 430
ORD010  | C014       | 425
```

Q7. Differentiate between list-wise deletion and pair-wise deletion.

Introduction: Both are deletion methods for treating missing data and are suitable only when the data is missing completely at random; otherwise they can bias the model.

Basis	List-wise Deletion	Pair-wise Deletion
Working	Removes an observation completely if any one of its variables is missing.	Uses every case in which the variables of interest are present and ignores only the missing pair.
Merit and demerit	Simple, but it reduces the sample size and the power of the model.	Keeps as many cases as possible, but different analyses use different sample sizes.

Q8. State the range of the correlation coefficient and interpret the values -1, 0 and +1.

Introduction: Correlation measures the strength and direction of the linear relationship between two continuous variables, and its value always lies between -1 and +1.

- **Values -1 and +1:** A value of +1 is a perfect positive linear correlation (both variables rise together) and -1 is a perfect negative linear correlation (one rises while the other falls). Values near these limits show a strong relationship.
- **Value 0 and formula:** A value of 0 means there is no linear correlation. It is computed as Covariance(X,Y) / SQRT(Var(X) × Var(Y)) and is available through CORREL() in Excel, PROC CORR in SAS and CORR() in SQL.

Q9. What is binning of variables? Give an example.

Introduction: Binning is a variable transformation technique that converts a variable into a few categories (bins) using its original values, percentiles or frequencies.

- **Purpose:** It simplifies implementation, because a separate strategy can be framed for each bin, and it reduces the influence of extreme values. The decision to bin is based on business understanding.
- **Example:** Sales agents can be grouped into age bins of below 30, 30 to 45 and above 45 years, or income can be grouped as Low, Average and High. In SQL, a CASE expression creates such bins.

Q10. What are dummy variables? Give an example.

Introduction: Dummy variables, also called indicator variables, are 0/1 variables created from a categorical variable so that it can be used as a numerical predictor in statistical models.

- **Working:** Each category becomes a separate column that holds 1 when the record belongs to that category and 0 otherwise. For a variable with n classes, n or n-1 dummy variables can be created.
- **Example:** The variable Gender produces Var_Male (1 for male, 0 for not male) and Var_Female (1 for female, 0 for not female), so a male employee is stored as Var_Male = 1 and Var_Female = 0.

Part B - 5 Marks

Q11. Explain the stages of data exploration.

Introduction: Data exploration is a sequence of stages that converts raw data from different sources into clean, well-understood data from which hypotheses for further analysis can be generated.

- **Data Collection and Cleaning:** Raw structured, semi-structured or unstructured data is gathered from SQL databases or spreadsheets. It is then cleaned by handling missing values, removing duplicates and correcting errors, because accurate analysis depends on a high-quality dataset.
- **Data Transformation:** The data is converted into suitable formats for analysis. This includes normalising values, creating new variables or features and sometimes applying algorithms to pre-process the data before it is analysed.
- **Data Visualization:** Analysts draw scatter plots, bar charts, histograms and box plots using tools such as Tableau, Excel and Python libraries. These visual representations make trends, patterns and outliers easy to identify.
- **Statistical Summary:** Basic statistics such as mean, median, mode and standard deviation summarise the data. Univariate and bivariate analyses are performed to understand each variable and the relationships between variables.
- **Hypothesis Generation:** Using insights from the earlier stages, analysts frame hypotheses and decide which areas need further study. This stage guides the subsequent data mining and machine learning workflows.

Q12. Explain how data exploration supports decision-making in business and industry. Illustrate your answer with a suitable SQL-based example.

Introduction: Data exploration supports decision-making by giving a clear picture of the data's structure, quality and patterns, so that business intelligence efforts are based on facts instead of assumptions.

- **Understanding the data:** Exploration reveals the structure, distribution and anomalies of a dataset. This foundational knowledge helps managers know what the data can and cannot answer before any decision is taken.
- **Identifying issues:** Early detection of outliers and missing values prevents them from distorting later results. A wrong sales figure found at this stage avoids a wrong stocking or pricing decision later.
- **Guiding analysis:** Key patterns and trends show which products, regions or customers deserve detailed analysis. This ensures that time, money and modelling effort are used efficiently.
- **Enhancing data quality:** Cleaning and transformation make the dataset more reliable, which leads to accurate analytical outcomes. Reports built on such data can be trusted by business users.
- **Facilitating decision-making:** Insights from exploration feed business intelligence and improve predictive modelling. Organisations can then plan inventory, marketing and pricing using evidence from their own data.

SQL-based example: A retailer explores the Sales(SaleID, Product, Region, Amount) table to compare regions.

```
SELECT   Region, COUNT(*) AS Orders,
         SUM(Amount) AS TotalSales,
         AVG(Amount) AS AvgSale
FROM     Sales
GROUP BY Region
ORDER BY TotalSales DESC;
```

```
Region | Orders | TotalSales | AvgSale
-------+--------+------------+--------
South  |   40   |   84000    |  2100
North  |   35   |   59500    |  1700
West   |   25   |   30000    |  1200
```

Decision: the South region performs best, so stock can be increased there, while the low average sale in the West calls for an investigation into pricing or promotions.

Q13. Differentiate between univariate analysis and bivariate analysis.

Introduction: Univariate analysis explores variables one at a time, whereas bivariate analysis studies the relationship between two variables at a pre-defined significance level.

Basis	**Univariate Analysis**	**Bivariate Analysis**
Number of variables	Examines a single variable at a time.	Examines two variables together.
Objective	Understands the central tendency, spread and distribution of one variable.	Finds association or disassociation between two variables.
Continuous variables	Mean, median, mode, range, quartiles, IQR, variance, standard deviation, skewness and kurtosis; histogram and box plot.	Scatter plot to see the pattern and correlation (-1 to +1) to measure its strength.
Categorical variables	Frequency table with Count and Count%; bar chart.	Two-way table, stacked column chart and chi-square test.

Basis	Univariate Analysis	Bivariate Analysis
Additional use	Highlights missing values and outliers in a variable.	Covers categorical-categorical, categorical-continuous and continuous-continuous combinations.

Example: Univariate analysis is SELECT AVG(Salary) FROM Employee; and bivariate analysis is SELECT Department, AVG(Salary) FROM Employee GROUP BY Department; which relates Salary to Department.

Q14. Differentiate between Z-test/T-test and ANOVA with a suitable example.

Introduction: Z-test, T-test and ANOVA are statistical tests used in bivariate analysis of a categorical and a continuous variable to decide whether differences between group means are statistically significant.

Basis	Z-test / T-test	ANOVA
Purpose	Tests whether the means of two groups are statistically different.	Tests whether the means of more than two groups are statistically different.
Number of groups	Exactly two groups (for example Male and Female).	Three or more groups (for example five exercise types).
Sample size	Z-test is used for large samples; T-test is used when each group has fewer than 30 observations.	Can be applied to groups of any size, such as 4 men per group.
Decision rule	A small probability value means the difference between the two averages is significant.	A small probability value means at least one group mean differs from the others.
Example	Compare the average Amount spent by male and female customers.	Test whether five exercise types change weight differently, using 20 men in 5 groups of 4.

T-test statistic: $t = (\bar{X}1 - \bar{X}2) / SQRT(S^2 \times (1/N1 + 1/N2))$, where S^2 is the pooled variance and the degrees of freedom are N1 + N2 - 2.

Q15. For a table Student(StudentID, Name, Department, Marks) in which some Marks are missing, write SQL queries to (i) list students with missing Marks, (ii) count the missing values, (iii) display Marks with missing values replaced by the overall average, (iv) display only complete records and (v) fill missing Marks with the department average.

Introduction: Missing values in SQL appear as NULL. They can be detected with IS NULL, removed by deletion or filled by imputation using the mean of the known values.

Sample data assumed (NULL means Marks is missing):

```
StudentID | Name   | Department | Marks
----------+--------+------------+------
S01       | Anil   | CSE        | 70
S02       | Bela   | CSE        | NULL
S03       | Chitra | ECE        | 80
S04       | Dev    | ECE        | NULL
S05       | Esha   | CSE        | 60
S06       | Farid  | ECE        | 90
```

- **(i) Students with missing Marks:** IS NULL filters the rows whose Marks value is unknown. The output lists S02 (Bela) and S04 (Dev).

```
SELECT * FROM Student WHERE Marks IS NULL;
```

- **(ii) Count of missing values:** COUNT(*) counts all rows, while COUNT(Marks) ignores NULLs, so their difference gives the number of missing values. The expected output is 2.

```
SELECT COUNT(*) - COUNT(Marks) AS MissingCount
FROM   Student;
```

- **(iii) Generalised imputation:** COALESCE returns the first non-NULL value, so every missing mark is replaced by the average of all known marks. AVG ignores NULLs, so the average is (70+80+60+90)/4 = 75, and S02 and S04 both show 75.

```
SELECT StudentID, Name,
       COALESCE(Marks,(SELECT AVG(Marks) FROM Student)) AS Marks
FROM   Student;
```

- **(iv) List-wise deletion:** Rows with missing Marks are excluded from the result, which is simple but reduces the sample size. Here the output has four rows: S01, S03, S05 and S06.

```
SELECT * FROM Student WHERE Marks IS NOT NULL;
```

- **(v) Similar case imputation:** Each missing mark is replaced by the average of the same Department. CSE average is (70+60)/2 = 65, so S02 becomes 65, and ECE average is (80+90)/2 = 85, so S04 becomes 85. In MySQL, use a join with a derived table instead of this correlated form.

```
UPDATE Student
SET    Marks = (SELECT AVG(s2.Marks) FROM Student s2
                WHERE s2.Department = Student.Department)
WHERE  Marks IS NULL;
```

Q16. Explain the types and causes of outliers with suitable examples.

Introduction: An outlier is an observation that appears far away from the overall pattern of a sample. Outliers are classified by the number of variables involved and by the reason for their occurrence, which decides how they are treated.

- **Univariate outliers:** These are found by looking at the distribution of a single variable. For example, in customer profiling, an average annual income of 0.8 million with two customers earning 4 and 4.2 million shows two clear outliers on a box plot.
- **Multivariate outliers:** These exist in an n-dimensional space and are visible only when two or more variables are viewed together. A height and a weight may each look normal on separate box plots, yet a scatter plot shows a few unusual height-weight combinations.
- **Artificial (error) causes:** Data entry errors occur when an income of 100,000 is typed as 1,000,000. Measurement errors, the most common source, occur when a faulty weighing machine gives high or low readings. Data processing errors arise during extraction from multiple sources.
- **Natural and situational causes:** Experimental error occurs when, in a 100 m sprint, one runner starts late and records a much higher time. Intentional outliers appear when teenagers under-report alcohol use, making honest values look extreme. Sampling error occurs when basketball players are included in a sample of athletes.
- **Impact on the dataset:** Outliers increase error variance, reduce the power of statistical tests, bias estimates and violate the assumptions of regression and ANOVA. For the values 4, 4, 5, 5, 5, 5, 6, 6, 6, 7, 7 the mean is 5.45, but adding one value of 300 raises the mean to 30.

Q17. Explain variable transformation. Describe the situations in which it is required and the common methods used.

Introduction: Variable transformation means replacing a variable with a function of it, such as its square root or logarithm, in order to change the distribution of the variable or its relationship with other variables.

- **Change of scale:** Transformation is a must when variables are measured on different scales, because standardising values makes them easier to compare and understand. Scaling changes the units but does not change the shape of the distribution.

- **Linearising relationships:** A linear relationship between variables is easier to model than a curved one. A scatter plot reveals the curve, and a transformation, usually the log, converts a non-linear relationship into a linear one and improves prediction.
- **Reducing skewness:** Symmetric distributions are easier to interpret, and some models need a normal distribution. A right-skewed variable is transformed by square root, cube root or log, while a left-skewed variable uses the square, cube or exponential.
- **Logarithm, square root and cube root:** The logarithm strongly reduces right skewness but cannot be applied to zero or negative values. Square root has a weaker effect and works for zero and positive values, whereas cube root also works for negative values and zero.
- **Binning for implementation:** Binning categorises a variable using original values, percentiles or frequencies. If performance rises with age, launching one program per age is impractical, so agents are grouped into below 30, 30 to 45 and above 45 and a strategy is framed for each group.

Q18. Explain the prediction-model and KNN methods of treating missing values with their advantages and disadvantages.

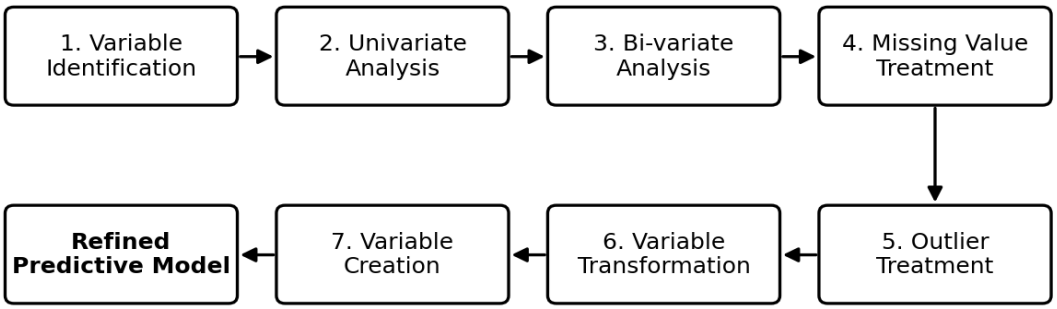
Introduction: Prediction-model and K-Nearest Neighbour (KNN) imputation are advanced methods that estimate missing values from the relationships present in the available data instead of using one fixed value.

- **Prediction model: working:** The dataset is divided into two sets: records without missing values become the training set and records with missing values become the test set. The attribute with missing values is treated as the target variable, a model is built from the other attributes, and its predictions fill the gaps.
- **Prediction model: drawbacks:** Regression, ANOVA, logistic regression and similar techniques can be used. However, the estimated values are usually more well-behaved than the actual values, and if the attributes have no relationship with the missing attribute, the estimates will not be precise.
- **KNN imputation: working:** A missing value is imputed using a chosen number (k) of the most similar records. Similarity is determined with a distance function, and the neighbours' values supply the estimate.
- **KNN advantages:** It can predict both qualitative and quantitative attributes, needs no separate predictive model for each attribute with missing data, treats attributes with many missing values easily and takes the correlation structure of the data into account.
- **KNN disadvantages:** It is very time-consuming on a large database because it searches all records for similar instances. The choice of k is critical: a high k includes records that differ greatly, while a low k misses significant ones.

Part C - 10 Marks

Q19. Explain the steps of data exploration and preparation for building a predictive model, with a suitable flow diagram.

Introduction: Data exploration and preparation is the systematic process of understanding, cleaning and enriching raw data before a predictive model is built, based on the principle that the quality of the inputs decides the quality of the output.



Steps 4 to 7 are repeated (iterated) several times before the model is finalised

- **Variable identification:** The predictor (input) and target (output) variables are identified first, followed by the data type (character or numeric) and the category (categorical or continuous) of each variable. In a student table predicting cricket play, Play_Cricket is the target and Gender, marks, height and weight are predictors.
- **Univariate analysis:** Variables are explored one by one. Continuous variables are summarised by central tendency (mean, median, mode, min, max) and spread (range, quartiles, IQR, variance, standard deviation, skewness), and categorical variables by frequency tables. It also highlights missing values and outliers.
- **Bi-variate analysis:** The relationship between two variables is examined at a pre-defined significance level. Scatter plots and correlation are used for two continuous variables, two-way tables and chi-square for two categorical variables, and box plots with Z-test, T-test or ANOVA for a categorical and a continuous variable.
- **Missing values treatment:** Missing data reduces the power of a model and may bias it. It is handled by list-wise or pair-wise deletion, mean, median or mode imputation (generalised or similar case), a prediction model or KNN imputation, depending on why the values are missing.
- **Outlier treatment:** Outliers are detected using box plots, histograms, scatter plots and rules such as 1.5 × IQR. They are then deleted, transformed, binned, imputed or treated separately, depending on whether the cause is an error or a natural variation.
- **Variable transformation:** Variables are replaced by functions such as log, square root or cube root, or are binned. This standardises scales, reduces skewness and converts non-linear relationships into linear ones so that the model works better.
- **Variable creation:** New variables are derived from existing ones, such as day, month and year from a date, dummy variables from a categorical variable, or a salutation extracted from a name. These highlight hidden relationships with the target variable.
- **Iteration:** Steps 4 to 7 are repeated several times before the refined model is developed, because treating missing values and outliers or creating new variables changes the data and can reveal new issues. Each pass improves the dataset.

- **Time and effort:** Data exploration, cleaning and preparation can take up to 70% of the total project time. Spending this effort once the business hypothesis is ready is worthwhile because errors left in the data spread into every later stage.
- **Outcome:** The result is a clean, well-understood dataset with identified variables, treated missing values and outliers and useful new features. This supports accurate predictive modelling, machine learning workflows and data-driven decision-making.

Q20. A retail company stores customer details in Customer(CustID, Gender, Age, Income, Purchased) and wants to predict whether a customer will purchase (Purchased = 1 or 0). Identify the variables and write SQL queries for the univariate analysis of continuous and categorical variables. Explain each query.

Introduction: Variable identification classifies every variable by its role, data type and category, and univariate analysis then explores each variable individually using methods that depend on whether it is categorical or continuous.

- **Predictor and target variables:** Gender, Age and Income are predictor (input) variables because they are used to explain the outcome. Purchased is the target (output) variable because the company wants to predict it. CustID only identifies a record and is not used for prediction.
- **Data type:** CustID and Gender are character data. Age, Income and Purchased are numeric. The data type tells us how a column can be stored and calculated in SQL.
- **Variable category:** Gender and Purchased are categorical variables because they hold a limited number of groups. Age and Income are continuous variables. This category decides the univariate method: frequency tables for categorical variables and summary statistics for continuous ones.
- **Sample data assumed:** The following eight rows are used to show the expected outputs of the queries in the next points.

```
CustID | Gender | Age | Income | Purchased
-------+--------+-----+--------+----------
C001   | M      | 28  | 45000  | 1
C002   | F      | 34  | 52000  | 0
C003   | M      | 45  | 61000  | 1
C004   | M      | 39  | 48000  | 0
C005   | F      | 25  | 39000  | 1
C006   | M      | 52  | 70000  | 1
C007   | F      | 31  | 43000  | 0
C008   | M      | 41  | 56000  | 1
```

- **Frequency table of Gender:** GROUP BY forms one group per category, COUNT(*) gives the Count and the division by the total number of rows gives Count%. Five of the eight customers are male, so the output shows M with 62.5% and F with 37.5%.

```
SELECT Gender, COUNT(*) AS Cnt,
       ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM Customer),1) AS CountPct
FROM   Customer
GROUP  BY Gender;

Gender | Cnt | CountPct
-------+-----+---------
M      |  5  |   62.5
F      |  3  |   37.5
```

- **Frequency table of the target:** The same query on Purchased shows how balanced the target variable is. Here, Purchased = 1 has a count of 5 (62.5%) and Purchased = 0 has a count of 3 (37.5%), so both outcomes are well represented for modelling.

```
SELECT Purchased, COUNT(*) AS Cnt
FROM   Customer
GROUP  BY Purchased;
```

- **Central tendency of Age:** AVG, MIN and MAX describe the centre and limits of a continuous variable. The average age is 295/8 = 36.875, the youngest customer is 25 and the oldest is 52.

```
SELECT AVG(Age) AS MeanAge, MIN(Age) AS MinAge, MAX(Age) AS MaxAge
FROM   Customer;

MeanAge | MinAge | MaxAge
--------+--------+-------
36.875  |   25   |   52
```

- **Spread of Age:** Range and standard deviation measure how widely the values are scattered around the mean. The range is 52 - 25 = 27 and the population standard deviation is about 8.51, so most customers lie within roughly 8.5 years of the mean.

```
SELECT MAX(Age) - MIN(Age) AS AgeRange,
       ROUND(STDDEV_POP(Age),2) AS StdDev
FROM   Customer;

AgeRange | StdDev
---------+-------
   27    |  8.51
```

- **Median and quartiles of Income:** PERCENTILE_CONT gives the median and quartiles, from which the IQR is found. For Income, the median is 50000, Q1 is 44500 and Q3 is 57250, so IQR = 12750. These values are used to detect outliers with the 1.5 × IQR rule.

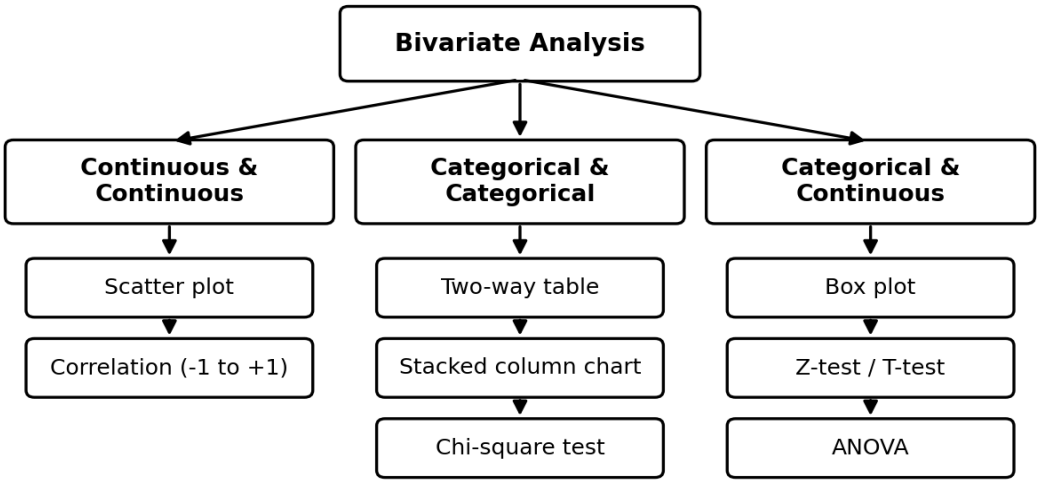
```
SELECT PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY Income) AS Q1,
       PERCENTILE_CONT(0.50) WITHIN GROUP (ORDER BY Income) AS Median,
       PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY Income) AS Q3
FROM   Customer;

Q1    | Median | Q3
------+--------+------
44500 | 50000  | 57250
```

- **Visualisation and data-quality checks:** A histogram and a box plot are drawn for Age and Income, and a bar chart for Gender and Purchased. Univariate analysis is also used to highlight missing and outlier values, such as a NULL Income or a very high Age, before further analysis.

Q21. An online store keeps monthly figures in Marketing(Month, AdSpend, Sales) and customer orders in Orders(OrderID, Gender, Purchased, Amount). Design a bivariate analysis for (i) two continuous variables, (ii) two categorical variables and (iii) a categorical and a continuous variable. Write SQL queries wherever possible and explain how the results are interpreted.

Introduction: Bivariate analysis finds the relationship between two variables, and the method used depends on whether the variables are categorical or continuous, as shown in the diagram below.



- **Scatter plot (continuous and continuous):** AdSpend is plotted on the X-axis and Sales on the Y-axis for each month. The pattern of the points shows whether the relationship is linear or non-linear, but a scatter plot does not tell us how strong the relationship is. For the six months below, the points rise almost along a straight line.

```
Month   | 1   | 2   | 3   | 4   | 5   | 6
--------+-----+-----+-----+-----+-----+----
AdSpend | 10  | 12  | 15  | 18  | 20  | 24
Sales   | 110 | 125 | 150 | 170 | 185 | 230
(AdSpend and Sales are in thousands)
```

- **Correlation (strength of the relationship):** Correlation = Covariance(X,Y) / SQRT(Var(X) × Var(Y)) and lies between -1 and +1. The SQL function CORR is available in PostgreSQL and Oracle; in other databases the formula is built from AVG.

```
SELECT ROUND(CORR(AdSpend, Sales)::numeric, 3) AS Correlation
FROM   Marketing;

Correlation
-----------
   0.995
```

- **Interpreting the correlation:** The covariance is 187.5, Var(AdSpend) is 22.58 and Var(Sales) is 1572.22, so the correlation is 187.5 / SQRT(22.58 × 1572.22) = 0.995. This is very close to +1, so higher advertising spend is associated with higher sales. It shows association, not proof of cause.
- **Two-way table (categorical and categorical):** A two-way table shows the count of each combination of Gender and Purchased, with one variable in rows and the other in columns. Assume 100 orders; from the counts, 60% of males purchased (30 of 50) compared with 30% of females (15 of 50).

```
SELECT Gender,
       SUM(CASE WHEN Purchased=1 THEN 1 ELSE 0 END) AS Yes,
       SUM(CASE WHEN Purchased=0 THEN 1 ELSE 0 END) AS No
FROM   Orders
GROUP  BY Gender;

Gender | Yes | No
-------+-----+----
M      | 30  | 20
F      | 15  | 35
```

- **Stacked column chart:** This is the visual form of the two-way table. Each bar represents one Gender and is divided into Purchased Yes and No segments, so a quick glance shows that the male bar has a larger purchase share than the female bar.
- **Chi-square test:** The test checks whether the relationship seen in the sample is strong enough to generalise to the population. It is based on the difference between observed (O) and expected (E) frequencies: $\chi^2 = \Sigma (O - E)^2 / E$, where E = row total × column total / sample size. A probability below 0.05 means the relationship is significant at 95% confidence.
- **Worked chi-square result:** Expected counts are 50 × 45 / 100 = 22.5 for Male-Yes and Female-Yes, and 50 × 55 / 100 = 27.5 for Male-No and Female-No. $\chi^2 = 56.25 \times (1/22.5 + 1/27.5 + 1/22.5 + 1/27.5) = 9.09$ with 1 degree of freedom, which exceeds 3.84, so Gender and Purchase are related. Cramer's V = SQRT(9.09/100) = 0.30 shows a moderate strength of relationship.
- **Box plot and group statistics (categorical and continuous):** A box plot of Amount is drawn for each Gender, and group-wise statistics are computed with GROUP BY. Box plots alone do not show significance, particularly when the groups are small, so a statistical test is used. Assume 50 orders per gender: M has an average Amount of 1850 and F has 1420.

```
SELECT Gender, COUNT(*) AS Orders, AVG(Amount) AS AvgAmount
FROM   Orders
GROUP  BY Gender;

Gender | Orders | AvgAmount
-------+--------+----------
M      |   50   |  1850.00
F      |   50   |  1420.00
```

- **Z-test and T-test:** These tests check whether the averages of two groups, here male and female Amount, differ significantly. If the probability is small, the difference between the two averages is significant. The T-test is used when the number of observations in each group is less than 30.
- **ANOVA:** ANOVA assesses whether the averages of more than two groups differ significantly, such as the average Amount across four regions. The group means come from a SQL GROUP BY query, and the test itself is run in a statistics tool such as Excel, Python or R.

Q22. A company stores employee data in Employee(EmpID, Gender, Experience, Salary) and many Salary values are missing. Explain why missing values must be treated, why they occur and how they can be handled using deletion and imputation methods, with SQL queries.

Introduction: Missing values are entries that are absent from the data, stored as NULL in SQL. Treating them is necessary because untreated missing data reduces the power and fit of a model and can lead to a biased model, wrong predictions or wrong classifications.

- **Why treatment is required:** Missing values prevent the behaviour of variables and their relationships from being analysed correctly. In a small student table, leaving two blank Gender entries untreated made males look more likely to play cricket than females. After the Gender values were filled in, females showed the higher chance (75% against 50%), so the conclusion reversed.
- **Causes: data extraction:** Problems in the extraction process can create missing values. They are usually easy to find and correct by double-checking the data with data guardians, and hashing procedures can be used to ensure correct extraction.
- **Causes: data collection:** These are harder to correct and are of four types. Missing completely at random: every record has the same chance of being missing (such as a coin toss deciding whether income is declared). Missing at random: the missing ratio varies with other input variables. Missing depending on unobserved predictors: a medical test causing discomfort increases drop-out. Missing depending on the value itself: very high or low earners decline to state income.
- **Detecting missing values:** Sample data is assumed below, where NULL means a missing Salary. COUNT(*) minus COUNT(Salary) gives the number of missing values, and the expected output is 2.

```
EmpID | Gender | Experience | Salary
------+--------+------------+-------
E1    | M      | 5          | 40000
E2    | F      | 3          | NULL
E3    | M      | 7          | 52000
E4    | M      | 4          | NULL
E5    | F      | 6          | 46000
E6    | M      | 8          | 58000
E7    | F      | 2          | 38000
E8    | M      | 5          | 42000

SELECT COUNT(*) - COUNT(Salary) AS MissingSalary FROM Employee;
```

- **List-wise deletion:** Every observation with any missing variable is deleted. It is simple, but it reduces the sample size and the power of the model, and it is safe only when data is missing completely at random. Here E2 and E4 are removed and six rows remain.

```
SELECT * FROM Employee WHERE Salary IS NOT NULL;
```

- **Pair-wise deletion:** Each analysis uses all cases in which the variables of interest are present. SQL aggregate functions ignore NULLs, so each average below uses its own available cases: Salary uses 6 rows and Experience uses 8 rows. This keeps more data, but the sample sizes differ between variables.

```
SELECT AVG(Salary) AS AvgSalary, AVG(Experience) AS AvgExp
FROM   Employee;

AvgSalary | AvgExp
----------+-------
46000.00  | 5.00
```

- **Generalised imputation:** The mean of all known values replaces each missing value. The known salaries add up to 276000 over 6 records, so the mean is 46000, and both E2 and E4 are shown with 46000.

```
SELECT EmpID, COALESCE(Salary,(SELECT AVG(Salary) FROM Employee)) AS Salary
FROM   Employee;
```

- **Similar case imputation:** The mean is calculated separately for each group, here Gender. The male average is (40000+52000+58000+42000)/4 = 48000 and the female average is (46000+38000)/2 = 42000. So E4 (male) is filled with 48000 and E2 (female) with 42000, which is more realistic than a single overall mean.

```
UPDATE Employee
SET    Salary = (SELECT AVG(e2.Salary) FROM Employee e2
                 WHERE e2.Gender = Employee.Gender)
WHERE  Salary IS NULL;
```

- **Prediction model:** Records with a known Salary form the training set and records with a missing Salary form the test set. Salary is treated as the target variable and is predicted from attributes such as Experience and Gender using regression or similar techniques. The estimates are usually too well-behaved, and the method fails if the attributes are unrelated to Salary.
- **KNN imputation and choice of method:** KNN fills a missing Salary from the k most similar employees found with a distance function; it is accurate but slow on large data, and the choice of k is critical. Deletion is suitable only for data missing completely at random, so for other types imputation is preferred to avoid bias.

Q23. A retail firm records order values in Orders(OrderID, CustomerID, Amount). Explain how outliers can be detected and treated, using SQL where possible, and describe the impact of an outlier on the dataset.

Introduction: An outlier is an observation that diverges from the overall pattern of a sample. It must be detected and treated carefully because it can drastically change the results of data analysis and statistical modelling.

- **Impact of an outlier:** Consider twelve orders: 410, 420, 380, 450, 400, 430, 390, 440, 415, 425, 405 and 9500 (ORD001 to ORD012). Without 9500, the mean is 415, the median is 415 and the standard deviation is 20.98. With it, the mean jumps to 1172.08, the median becomes 417.5 and the standard deviation rises to 2622.69. This shows that the mean and standard deviation are badly distorted, whereas the median is almost unaffected.
- **Detection by visualisation:** The most common method is visual: a box plot shows points beyond the whiskers, a histogram shows isolated bars far from the rest, and a scatter plot reveals unusual combinations of two variables. Here, a box plot of Amount would show ORD012 floating far above the box.
- **Detection by 1.5 × IQR rule:** Any value below Q1 - 1.5 × IQR or above Q3 + 1.5 × IQR is an outlier. For the data, Q1 = 403.75, Q3 = 432.5 and IQR = 28.75, so the limits are 360.63 and 475.63, and only ORD012 (9500) lies outside them.

```
WITH q AS (
  SELECT PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY Amount) AS q1,
```

```
         PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY Amount) AS q3
  FROM Orders)
SELECT o.OrderID, o.Amount
FROM   Orders o, q
WHERE  o.Amount < q.q1 - 1.5*(q.q3 - q.q1)
   OR  o.Amount > q.q3 + 1.5*(q.q3 - q.q1);

OrderID | Amount
--------+-------
ORD012  | 9500
```

- **Detection by three standard deviations:** Data points three or more standard deviations from the mean are treated as outliers. The mean is 1172.08 and the standard deviation is 2622.69, so the upper limit is about 9040 and ORD012 is flagged. Because the outlier itself inflates the standard deviation, this rule is less sensitive than the IQR rule.

```
SELECT OrderID, Amount
FROM   Orders
WHERE  ABS(Amount - (SELECT AVG(Amount) FROM Orders))
       > 3 * (SELECT STDDEV(Amount) FROM Orders);
```

- **Capping and multivariate detection:** Capping treats any value outside the 5th to 95th percentile range as an outlier. Bivariate and multivariate outliers are measured by influence, leverage or distance, using indices such as Mahalanobis' distance and Cook's D. In SAS, PROC UNIVARIATE and PROC SGPLOT are used.
- **Finding the cause first:** The ideal way to tackle an outlier is to find out why it occurred. An artificial outlier (data entry, measurement or processing error) can be corrected or removed, whereas a natural outlier, such as a genuine bulk order, is real information and should not be thrown away blindly.
- **Deleting observations:** Outliers caused by data entry or processing errors, or very few in number, are deleted, and trimming at both ends can also be used. The query keeps only values within the IQR limits; the average then returns to 415.00.

```
SELECT AVG(Amount) AS AvgAmount
FROM   Orders
WHERE  Amount BETWEEN 360.63 AND 475.63;
```

- **Transforming and binning:** Taking the natural log reduces the variation caused by extreme values: LN(9500) = 9.159 compared with LN(450) = 6.109, so the outlier is about 1.5 times larger instead of about 21 times. Binning (as done by decision trees) also limits its effect, and weights can be assigned to observations.

```
SELECT OrderID, Amount, ROUND(LN(Amount)::numeric,3) AS LogAmount
FROM   Orders;
```

- **Imputing:** If the outlier is artificial, it can be imputed in the same way as a missing value, using the mean, median or mode, or a value predicted by a statistical model. For example, ORD012 can be replaced by the median 415 of the remaining orders.

```
UPDATE Orders SET Amount = 415 WHERE OrderID = 'ORD012';
```

- **Treating separately:** If there are many outliers, they should not be deleted or altered one by one. Both groups are treated as separate entities, an individual model is built for each group and the outputs are then combined. This keeps genuine extreme customers, such as wholesale buyers, in the analysis.

Q24. An e-commerce business stores data in Orders(OrderID, Gender, CustAge, OrderDate, Amount). Explain feature engineering and design SQL queries for variable transformation and variable creation (log transformation, binning, date features and dummy variables). State the purpose of each query.

Introduction: Feature engineering is the science and art of extracting more information from existing data. No new data is added; the existing data is made more helpful for modelling.

- **Meaning and example:** Suppose footfall in a shopping mall is predicted from dates. Footfall is affected more by the day of the week than by the day of the month, but this information is hidden in the date. Bringing it out as a new variable improves the model.
- **Process:** Feature engineering is done after the first five steps of data exploration: variable identification, univariate analysis, bivariate analysis, missing value treatment and outlier treatment. It has two parts, variable transformation and variable (feature) creation. The following sample data is used for the outputs.

```
OrderID | Gender | CustAge | OrderDate  | Amount
--------+--------+---------+------------+-------
ORD001  | M      | 27      | 2026-09-21 | 100
ORD002  | F      | 38      | 2026-10-03 | 1000
ORD003  | M      | 52      | 2026-11-01 | 100000
```

- **Variable transformation: when required:** Transformation is used to change or standardise the scale, to turn a non-linear relationship into a linear one, to reduce skewness so that the distribution becomes closer to symmetric, and to make results easier to implement through binning.
- **Log transformation:** The natural log compresses large values and reduces right skewness, and it makes a curved relationship more linear. The query adds a new column LogAmount. It cannot be used for zero or negative values, so a WHERE Amount > 0 condition guards it.

```
SELECT OrderID, Amount, ROUND(LN(Amount)::numeric,4) AS LogAmount
FROM   Orders WHERE Amount > 0;

OrderID | Amount | LogAmount
--------+--------+----------
ORD001  | 100    | 4.6052
ORD002  | 1000   | 6.9078
ORD003  | 100000 | 11.5129
```

- **Square root and cube root:** These have a weaker effect than the logarithm but still reduce right skewness. Square root works for zero and positive values; cube root also works for negative values, so it is calculated with the sign handled separately. Here SQRT(100) = 10, SQRT(1000) = 31.62 and SQRT(100000) = 316.23.

```
SELECT OrderID, ROUND(SQRT(Amount)::numeric,2) AS SqrtAmount
FROM   Orders;
```

- **Binning with CASE:** Binning converts the continuous CustAge into three categories, so that a separate strategy can be formulated for each age group. The expected output is ORD001 as Below 30, ORD002 as 30-45 and ORD003 as Above 45.

```
SELECT OrderID, CustAge,
       CASE WHEN CustAge < 30 THEN 'Below 30'
            WHEN CustAge <= 45 THEN '30-45'
            ELSE 'Above 45' END AS AgeGroup
FROM   Orders;
```

- **Variable creation: date features:** New variables such as day, month, year and weekday are extracted from OrderDate because they may have a better relationship with the target. Here the weekday code is 1 for Monday, 6 for Saturday and 0 for Sunday, and the output for ORD001 is day 21, month 9, year 2026, weekday 1.

```
SELECT OrderID,
       EXTRACT(DAY   FROM OrderDate) AS New_Day,
       EXTRACT(MONTH FROM OrderDate) AS New_Month,
       EXTRACT(YEAR  FROM OrderDate) AS New_Year,
       EXTRACT(DOW   FROM OrderDate) AS WeekDay
FROM   Orders;
```

- **Dummy variables:** Gender is converted into numerical indicator columns Var_Male and Var_Female so that it can be used as a predictor in statistical models. Output: ORD001 gets Var_Male = 1 and Var_Female = 0, and ORD002 gets Var_Male = 0 and Var_Female = 1.

```
SELECT OrderID,
       CASE WHEN Gender='M' THEN 1 ELSE 0 END AS Var_Male,
       CASE WHEN Gender='F' THEN 1 ELSE 0 END AS Var_Female
FROM   Orders;
```

- **Derived variables:** A derived variable is created from existing variables using functions or other methods. In the Titanic data, the salutation (Master, Mr, Miss, Mrs) extracted from the name was used to predict missing ages. Which variable to create depends on the analyst's business understanding, curiosity and hypotheses.
- **Benefits of feature engineering:** It highlights hidden relationships in the data, such as weekday effects, and gives the model more useful inputs without collecting new data. Variable transformation and creation therefore have a remarkable impact on prediction and on the accuracy of the final model.